

Reduction of finite-size effects for second-order Møller–Plesset perturbation theory with singularity subtraction

Stephen Jon Quiton¹, Juan D. F. Pottecher¹, Martin Head-Gordon^{*1}, and Lin Lin^{*2,3}

¹Department of Chemistry, University of California, Berkeley, California 94720, United States

²Department of Mathematics, University of California, Berkeley, California 94720, United States

³Applied Mathematics and Computational Research Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720, United States

*Email: mhg@cchem.berkeley.edu, linlin@math.berkeley.edu

Abstract

Second-order Moller-Plesset perturbation theory (MP2) provides accurate correlation energies for periodic systems but suffers from finite-size errors (FSEs) that have inverse volume scaling due to the Coulomb kernel singularity in reciprocal space. This error scaling limits the routine applicability of MP2 to real materials, requiring prohibitively dense $\mathbf{k}$ -point meshes for convergence toward the thermodynamic limit (TDL). We introduce MP2 singularity subtraction (MP2SS), a systematic approach that applies the singularity subtraction strategy to reduce MP2 FSEs. The method employs auxiliary functions and fitting procedures that consider both the singularities present at the origin in reciprocal space and also the discontinuities in the MP2 structure factor that arise from finite $\mathbf{k}$ -point sampling. We present three possible MP2SS configurations (Gaussian, exponential, and tuned) which use different combinations of decay functions and demonstrate their performance for gapped systems. All MP2SS configurations consistently achieve millihartree accuracy for correlation energies at coarser $\mathbf{k}$ -point meshes than with no correction. Our results establish singularity subtraction as a powerful and flexible approach for mitigating finite-size errors in periodic correlation methods and provide a foundation for extending the technique to higher-order perturbation theories and other post-SCF methods.

1 Introduction

Wavefunction-based methods, such as second-order Moller-Plesset Perturbation theory (MP2) and coupled-cluster (CC), have garnered recent attention in the periodic quantum chemistry literature [1–8], owing both to their effectiveness in describing certain materials and to increased compute capabilities in recent years. These *ab initio* methods offer a systematically improvable suite of approaches to recover the electron correlation energy by correcting the mean-field Hartree-Fock (HF) wavefunction. Periodic MP2, which will be the focus of this work, has been shown to provide promising results across a broad range of crystals, especially those dominated by dispersion interactions [1, 3, 6, 9, 10]. That being said, MP2 theory has its own deficiencies, as it is ill-suited for describing small-gapped systems and results in a diverging correlation energy for metals. Furthermore, MP2 is also known to overestimate some classes of dispersion interactions such as π -stacking [11]. Finally, just as in periodic HF, periodic MP2 suffers from finite-size effects, which constitute part of the deterrence against its widespread use in solid-state quantum chemistry [12, 13].

Finite-size errors (FSEs) are errors in computed values that come from the $\mathbf{k}$ -point discretization of integrals over reciprocal space. These errors are fundamental and specific to periodic quantum chemistry calculations, independent of basis set incompleteness (BSIE) and correlation errors that are also encountered in molecular calculations [13, 14]. To eliminate FSEs, reciprocal space must be sampled with a sufficiently

fine $\mathbf{k}$ -point mesh in the First Brillouin Zone (FBZ). Without any corrections and assuming all orbitals $\{\psi_{i\mathbf{k}}\}$ and their energies $\{\epsilon_{i\mathbf{k}}\}$ can be computed exactly for any $\mathbf{k}$, the FSE for the MP2 energy scales as $\mathcal{O}(N_{\mathbf{k}}^{-1})$, where $N_{\mathbf{k}}$ is the number of $\mathbf{k}$ -points sampled in the FBZ. This error originates primarily from an excluded term where the Coulomb kernel is singular in reciprocal space [13]. Several methods exist in the literature to address the FSE specifically for MP2, including the staggered mesh method [12, 13, 15] and cubic splines interpolation [16]. The staggered mesh method, while simple in concept and implementation, incurs additional memory and compute costs owing to the calculation of the virtual space at the shifted set of $\mathbf{k}$ -points. Cubic splines interpolation has also been demonstrated in its effectiveness for several unit cells and is easily extendable to higher order post-HF methods [16], yet its success is dependent on the convergence of the MP2 structure factor with respect to $N_{\mathbf{k}}$. However, at finite $N_{\mathbf{k}}$, there are discontinuities in the structure factor for both the direct and exchange terms of MP2, even if the densities and orbital energies are themselves fully converged [12, 13]. These discontinuities limit the accuracy of interpolation-based methods in correcting the FSE, and thus, there is an opportunity to achieve even faster convergence to the TDL by considering the reciprocal space singularities in the MP2 energy.

In this work, we introduce the systematic treatment of the MP2 FSE with singularity subtraction, which has previously been successfully employed for exact exchange in periodic HF and hybrid DFT [17, 18]. This strategy seeks to address the finite-size error by finding an auxiliary function that satisfies two criteria: (1) it has a singularity matching or closely resembling that of the original integrand, and (2) its integration is a closed-form expression or easily computable to an arbitrarily high accuracy. Similarly to exact-exchange singularity subtraction (ExxSS), MP2 singularity subtraction (MP2SS) involves first computing the structure factor for the MP2 correlation energy and then fitting the auxiliary function to that structure factor. For MP2, obtaining an appropriate auxiliary function requires more care, as the structure factor itself has singularities, and those singularities manifest differently in the direct and exchange terms.

The rest of the paper is laid out as follows. In section 2, we review MP2 for periodic systems, including its reciprocal space singularities, and singularity subtraction. Then, in section 3, we outline the MP2SS strategy for each singularity type and present an overall procedure for real systems. Finally, in section 4, we demonstrate MP2SS's ability to recover correlation energies for a test set of real materials. As MP2 diverges for metals and performs sub-optimally for small-gap systems, our study is currently limited to insulating systems.

2 Theoretical Background

2.1 Periodic MP2 Theory

We work in the Bloch-orbital framework, using i, j, k, l to label occupied crystalline/molecular orbitals (MOs) and a, b, c, d to label virtual MOs. We denote by $N_{\mathbf{k}}$ the total number of $\mathbf{k}$ -points in the Monkhorst-Pack mesh [19], and by Ω the unit cell, with volume $|\Omega|$. Its corresponding FBZ is Ω^* , with volume $|\Omega^*|$. Throughout the paper, $\mathbf{R} \in \mathbb{L}$ denotes a real-space Bravais-lattice vector and $\mathbf{G} \in \mathbb{L}^*$ denotes a reciprocal-lattice vector. A Bloch orbital with band index i and crystal momentum $\mathbf{k}$ is written as $\psi_{i\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N_{\mathbf{k}}}} e^{-i\mathbf{k} \cdot \mathbf{r}} \phi_{i\mathbf{k}}(\mathbf{r})$, where $\phi_{i\mathbf{k}}(\mathbf{r})$ is the real-space periodic part of the MO. The corresponding pair density is defined as

$$\rho_{i\mathbf{k}_i j\mathbf{k}_j}(\mathbf{r}) = \phi_{i\mathbf{k}_i}^*(\mathbf{r}) \phi_{j\mathbf{k}_j}(\mathbf{r}) := \frac{1}{|\Omega|} \sum_{\mathbf{G} \in \mathbb{L}^*} \hat{\rho}_{i\mathbf{k}_i j\mathbf{k}_j}(\mathbf{G}) e^{i\mathbf{G} \cdot \mathbf{r}}, \quad (1)$$

where $\hat{\rho}_{i\mathbf{k}_i j\mathbf{k}_j}(\mathbf{G})$ denotes the Fourier coefficient of the MO product.

With this definition, two-electron integrals in the Bloch orbital basis can be expressed as

$$\langle i\mathbf{k}_i, j\mathbf{k}_j | a\mathbf{k}_a, b\mathbf{k}_b \rangle = \frac{1}{|\Omega^s|} \sum'_{\mathbf{G} \in \mathbb{L}^*} \hat{\rho}_{i\mathbf{k}_i a\mathbf{k}_a}(\mathbf{G}) \hat{V}(\mathbf{q} + \mathbf{G}) \hat{\rho}_{j\mathbf{k}_j b\mathbf{k}_b}(\mathbf{G}_{\mathbf{k}_i, \mathbf{k}_j}^{\mathbf{k}_a, \mathbf{k}_b} - \mathbf{G}). \quad (2)$$

Here $|\Omega^s| = N_{\mathbf{k}}|\Omega|$ denotes the volume of the periodic Born-von Karman (BvK) supercell corresponding to the discretized Brillouin zone, $\mathbf{G}_{\mathbf{k}_i, \mathbf{k}_j}^{\mathbf{k}_a, \mathbf{k}_b} := \mathbf{k}_i + \mathbf{k}_j - \mathbf{k}_a - \mathbf{k}_b \in \mathbb{L}^*$ by crystal momentum conservation, $\mathbf{q} = \mathbf{k}_a - \mathbf{k}_i$ is the momentum-transfer vector, $\sum'$ indicates that the (singular) term where $\mathbf{q} + \mathbf{G} = \mathbf{0}$ is excluded, and $\hat{V}(\mathbf{q} + \mathbf{G}) = 4\pi/|\mathbf{q} + \mathbf{G}|^2$ is the Coulomb kernel in reciprocal space. If $\mathbf{k}_i, \mathbf{k}_a$ belong to a grid

in the FBZ denoted by $\mathcal{K}$, then $\mathbf{q}$ belongs to an origin-centered grid denoted by $\mathcal{K}_{\mathbf{q}}$. The tessellation of $\mathcal{K}_{\mathbf{q}}$ across the whole domain of $\mathbb{R}^3$ is denoted by $\mathcal{K}_{\mathbf{q}} + \mathbb{L}^* := \{\mathbf{q} + \mathbf{G} \mid \mathbf{q} \in \mathcal{K}_{\mathbf{q}}, \mathbf{G} \in \mathbb{L}^*\}$.

The restricted-orbital closed shell MP2 correlation energy at the finite $N_{\mathbf{k}}$ can first be expressed as a triple quadrature over the FBZ of the regular MP2 expression

$$E_{\text{MP2}}(N_{\mathbf{k}}) = \frac{1}{N_{\mathbf{k}}} \sum_{ijab} \sum_{\mathbf{k}_i \mathbf{k}_j \mathbf{k}_a \in \mathcal{K}} \frac{2\langle i\mathbf{k}_i, j\mathbf{k}_j | a\mathbf{k}_a, b\mathbf{k}_b \rangle - \langle i\mathbf{k}_i, j\mathbf{k}_j | b\mathbf{k}_b, a\mathbf{k}_a \rangle}{\varepsilon_{i\mathbf{k}_i, j\mathbf{k}_j}^{a\mathbf{k}_a, b\mathbf{k}_b}} \langle a\mathbf{k}_a, b\mathbf{k}_b | i\mathbf{k}_i, j\mathbf{k}_j \rangle, \quad (3)$$

where $\mathbf{k}_b$ is determined by crystal momentum conservation, and the energy denominator is

$$\varepsilon_{i\mathbf{k}_i, j\mathbf{k}_j}^{a\mathbf{k}_a, b\mathbf{k}_b} := \varepsilon_{i\mathbf{k}_i} + \varepsilon_{j\mathbf{k}_j} - \varepsilon_{a(\mathbf{k}_i + \mathbf{q})} - \varepsilon_{b(\mathbf{k}_j - \mathbf{q})}. \quad (4)$$

The TDL is achieved as the mesh $\mathcal{K}$ becomes dense in Ω^* and each Riemann sum $\frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{k} \in \mathcal{K}}$ converges to the corresponding integral $\frac{1}{|\Omega^*|} \int_{\Omega^*} d\mathbf{k}$.

Our discussion on FSEs for MP2 focuses on the part introduced by the quadratures involving the Coulomb-kernel singularity. Previous analyses [12, 13] show that, for insulating systems and in the absence of additional non-smoothness from the orbital energies or amplitudes, this is the leading contribution to the inverse-volume scaling of the error. Additional errors arising from the orbitals $\{\psi_{i\mathbf{k}}\}$, the orbital energies $\{\epsilon_{i\mathbf{k}}\}$, and basis-set incompleteness, while important, are not considered in this work.

2.2 Classification of the MP2 Singularities

The direct term of the MP2 energy (the first term in the numerator of (3)) at the TDL can be expressed as an integral over the FBZ.

$$E_{\text{MP2,d}}^{\text{TDL}} = \frac{1}{|\Omega^*|} \int_{\Omega^*} d\mathbf{q} \sum_{\mathbf{G}, \mathbf{G}' \in \mathbb{L}^*} \frac{S_{\text{d}}(\mathbf{q}, \mathbf{G}, \mathbf{G}')}{|\mathbf{q} + \mathbf{G}|^2 |\mathbf{q} + \mathbf{G}'|^2}, \quad (5)$$

where $S_{\text{d}}(\mathbf{q}, \mathbf{G}, \mathbf{G}')$ is the direct term structure factor, defined as

$$S_{\text{d}}(\mathbf{q}, \mathbf{G}, \mathbf{G}') = \frac{2}{|\Omega^*|^2} \int_{\Omega^*} d\mathbf{k}_i \int_{\Omega^*} d\mathbf{k}_j \sum_{ijab} \frac{r_{ijab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{q} + \mathbf{G}) r_{ijab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{q} + \mathbf{G}')^*}{\varepsilon_{i\mathbf{k}_i, j\mathbf{k}_j}^{a\mathbf{k}_a, b\mathbf{k}_b}}, \quad (6)$$

and $r_{ijab}(\mathbf{k}_i, \mathbf{k}_j, \mathbf{q} + \mathbf{G}) = \frac{4\pi}{|\Omega|} \hat{\rho}_{i\mathbf{k}_i a(\mathbf{k}_i + \mathbf{q})}(\mathbf{G}) \hat{\rho}_{j\mathbf{k}_j b(\mathbf{k}_j - \mathbf{q})}(-\mathbf{G})$ is the product of the Fourier transforms of two MO products. Note that in this definition, which utilizes $\mathbf{q}$ instead of $\mathbf{k}_a$ and $\mathbf{k}_b$, crystal momentum conservation holds since $\mathbf{G}_{\mathbf{k}_i, \mathbf{k}_j}^{\mathbf{k}_a, \mathbf{k}_b} = \mathbf{G}_{\mathbf{k}_i, \mathbf{k}_j}^{\mathbf{k}_i + \mathbf{q}, \mathbf{k}_j - \mathbf{q}} = \mathbf{0}$. Alternatively, with $\Delta\mathbf{G} = \mathbf{G}' - \mathbf{G}$ and taking advantage of the symmetry of the direct structure factor, $S_{\text{d}}(\mathbf{q} + \mathbf{G}, \mathbf{0}, \mathbf{G}' - \mathbf{G}) = S_{\text{d}}(\mathbf{q}, \mathbf{G}, \mathbf{G}')$, we can rewrite the MP2 direct energy as

$$E_{\text{MP2,d}}^{\text{TDL}} = \frac{1}{|\Omega^*|} \int_{\mathbb{R}^3} d\mathbf{q} \sum_{\Delta\mathbf{G} \in \mathbb{L}^*} \frac{S_{\text{d}}(\mathbf{q}, \mathbf{0}, \Delta\mathbf{G})}{|\mathbf{q}|^2 |\mathbf{q} + \Delta\mathbf{G}|^2} = \frac{1}{|\Omega^*|} \int_{\mathbb{R}^3} d\mathbf{q} \left[\frac{S_{\text{d}}(\mathbf{q}, \mathbf{0}, \mathbf{0})}{|\mathbf{q}|^4} + \sum_{\Delta\mathbf{G} \neq \mathbf{0}} \frac{S_{\text{d}}(\mathbf{q}, \mathbf{0}, \Delta\mathbf{G})}{|\mathbf{q}|^2 |\mathbf{q} + \Delta\mathbf{G}|^2} \right] \quad (7)$$

This reduction from an FBZ integral with two reciprocal-lattice sums to an integral over $\mathbb{R}^3$ turns $N_{\mathbf{G}}^2$ singularity-subtraction problems over Ω^* into $N_{\mathbf{G}}$ problems over all of reciprocal space. From eq (7) we observe two types of singularities. The first is a fourth-order (quartic) singularity at $\Delta\mathbf{G} = \mathbf{0}$ as $\mathbf{q} \rightarrow \mathbf{0}$, represented by the first term. The second is a collection of second-order (quadratic) singularities when $\Delta\mathbf{G} \neq \mathbf{0}$, occurring as $\mathbf{q} \rightarrow \mathbf{0}$ or $\mathbf{q} \rightarrow -\Delta\mathbf{G}$. Under the same smoothness and gap assumptions stated above, both types of singularities produce $\mathcal{O}(N_{\mathbf{k}}^{-1})$ contributions to the MP2 FSE.

Analogously, the exchange part of the MP2 correlation energy (i.e. arising from the second term in the numerator of eq (3)) at the TDL can be expressed as a double integral over the FBZ:

$$E_{\text{MP2,x}}^{\text{TDL}} = \frac{1}{|\Omega^*|^2} \int_{\Omega^*} d\mathbf{q}_1 \int_{\Omega^*} d\mathbf{q}_2 \sum_{\mathbf{G}, \mathbf{G}' \in \mathbb{L}^*} \frac{S_{\text{x}}(\mathbf{q}_1, \mathbf{q}_2, \mathbf{G}, \mathbf{G}')}{|\mathbf{q}_1 + \mathbf{G}|^2 |\mathbf{q}_2 + \mathbf{G}'|^2}, \quad (8)$$

where the unfolded exchange structure factor is defined as

$$S_x(\mathbf{q}_1, \mathbf{q}_2, \mathbf{G}, \mathbf{G}') = -\frac{1}{|\Omega^*|} \int_{\Omega^*} d\mathbf{k}_i \sum_{ijab} \frac{r_{ijba}(\mathbf{k}_i, \mathbf{k}_i + \mathbf{q}_1 + \mathbf{q}_2, \mathbf{q}_1 + \mathbf{G}) r_{abij}(\mathbf{k}_i + \mathbf{q}_2, \mathbf{k}_i + \mathbf{q}_1, -\mathbf{q}_2 - \mathbf{G}')}{\varepsilon_{i\mathbf{k}_i, j(\mathbf{k}_i + \mathbf{q}_1 + \mathbf{q}_2)}^{a(\mathbf{k}_i + \mathbf{q}_2), b(\mathbf{k}_i + \mathbf{q}_1)}}. \quad (9)$$

Above, we use the convention $\mathbf{q}_2 = \mathbf{k}_a - \mathbf{k}_i$, $\mathbf{q}_1 = \mathbf{k}_b - \mathbf{k}_i$, and therefore by crystal momentum conservation, $\mathbf{k}_j = \mathbf{k}_i + \mathbf{q}_1 + \mathbf{q}_2$. Note that because the sums over the reciprocal lattice are taken over all of $\mathbb{L}^*$, the exchange structure factor obeys $S_x(\mathbf{q}_1, \mathbf{q}_2, \mathbf{G}, \mathbf{G}') = S_x(\mathbf{q}_1 + \mathbf{G}, \mathbf{q}_2 + \mathbf{G}', \mathbf{0}, \mathbf{0})$. In other words, the structure factor can be re-expressed to depend only on two independent wavevectors. Consequently, eq (8) can be rewritten as a double integral over all of reciprocal space:

$$E_{\text{MP2},x}^{\text{TDL}} = \frac{1}{|\Omega^*|^2} \int_{\mathbb{R}^3} d\mathbf{q}_1 \int_{\mathbb{R}^3} d\mathbf{q}_2 \frac{S_x(\mathbf{q}_1, \mathbf{q}_2, \mathbf{0}, \mathbf{0})}{|\mathbf{q}_1|^2 |\mathbf{q}_2|^2}, \quad (10)$$

The finite-size error in the exchange term when discretizing over $\mathbf{q}_1$ and $\mathbf{q}_2$ comes from the singularities at $\mathbf{q}_1 = \mathbf{q}_2 = \mathbf{0}$. Under the same regularity assumptions used for the direct term, this contribution is also $\mathcal{O}(N_k^{-1})$ [12, 13].

2.3 Numerical Quadrature and Singularity Subtraction (SS)

Uniform quadrature is the simplest numerical integration technique, in which the integrand is discretized into equally-sized pieces over a uniform grid, and the integral is approximated by summing these pieces with equal weighting. Most periodic electronic structure codes are implicitly using uniform quadrature when they use uniform k-grids to sample the FBZ.

There are three aspects of an integral that determine the performance of an approximation by uniform quadrature: (1) the domain of integration, (2) the periodicity of the integrand, and (3) the regularity of the integrand (e.g., how many continuous derivatives it has). For our purposes, a uniform quadrature on a regular grid is effective in the following two settings [20]:

- (A) If the domain of integration is the full space $\mathbb{R}^d$, then convergence can be rapid provided the integrand is sufficiently smooth and decays sufficiently fast at infinity.
- (B) If the domain of integration is a d -dimensional parallelepiped in $\mathbb{R}^d$, then convergence can be rapid provided the integrand is periodic on that domain and sufficiently smooth.

In both cases, higher regularity improves the asymptotic rate, but the precise rate depends on the available smoothness; with only finitely many derivatives one generally expects algebraic convergence, whereas exponential convergence requires stronger analyticity assumptions. Thus, one should avoid employing uniform quadrature on integrands that are not periodic on a finite domain of integration or on integrands that contain singularities or cusps, since these features degrade the regularity needed for rapid convergence and can dominate the discretization error.

Singularity subtraction (SS)[21, 22] is a technique that reduces quadrature errors of singular integrands by adding and subtracting an auxiliary function that reproduces the leading local singular behavior of the integrand [17, 18, 23, 24]. The goal is not to remove the singular contribution from the problem, but to isolate it into a term that can be integrated analytically or to very high accuracy, leaving behind a smoother remainder that is better suited to uniform quadrature. For example, if one is trying to compute the integral over $\mathbb{R}^d$ of a function $f(\mathbf{x})$ that has an isolated integrable singularity at $\mathbf{x} = \mathbf{0}$ with uniform quadrature (on an origin-centered grid), then the naive approach would be to simply exclude the singular point and apply quadrature to the rest of the domain, i.e.,

$$\int_{\mathbb{R}^d} f(\mathbf{x}) d\mathbf{x} \approx \sum_{i=1}^M{}' w(\mathbf{x}_i) f(\mathbf{x}_i), \quad (11)$$

where the primed sum means the singular point is excluded, and $w(\mathbf{x}_i)$ is the quadrature weight at the node i . The singularity and the excluded quadrature point hinder convergence towards the true value of the integral as the number of quadrature points increases.

The SS method accelerates this convergence by introducing an auxiliary function $g(\mathbf{x})$, resulting in the following splitting of the integrand

$$\int_{\mathbb{R}^d} f(\mathbf{x}) d\mathbf{x} = \int_{\mathbb{R}^d} (f(\mathbf{x}) - g(\mathbf{x})) d\mathbf{x} + \int_{\mathbb{R}^d} g(\mathbf{x}) d\mathbf{x} \approx \sum_i' w(\mathbf{x}_i) (f(\mathbf{x}_i) - g(\mathbf{x}_i)) + \int_{\mathbb{R}^d} g(\mathbf{x}) d\mathbf{x}, \quad (12)$$

Here, $g(\mathbf{x})$ should be chosen so that the second term (the integral involving g) has a closed-form expression or is easily computable, and $g(\mathbf{x})$ should reproduce the leading local asymptotic behavior of $f(\mathbf{x})$ near $\mathbf{x} = \mathbf{0}$. This improves the asymptotic convergence rate for the quadrature of $f - g$. If, in addition, $g(\mathbf{x})$ also matches $f(\mathbf{x})$ well away from the singularity, then the quadrature term $\sum_i w(\mathbf{x}_i) (f(\mathbf{x}_i) - g(\mathbf{x}_i))$ is also small, and this reduces the error prefactor. Just as in the case with HF exact exchange, good performance in the pre-asymptotic regime is critical to the success of SS schemes when applied to the MP2 correlation energy, and, as will be discussed at length in this work, a careful selection of auxiliary function is required to systematically treat the singularities identified above. The reformulation of the MP2 correlation energy in eqs (7) and (10) allows us to apply SS in the setting (A) where the domain of integration is $\mathbb{R}^d$, which will be discussed in more detail next.

3 Singularity Subtraction Applied to Periodic MP2

We now apply SS to the MP2 correlation energy. Unlike HF exact exchange, MP2 has multiple types of singularities in reciprocal space as expressed in eqs (7) and (10). Each singularity type needs special care in terms of auxiliary function design and fitting strategy. The exchange term singularities, while still important to correct, are not expected to contribute as much to the FSE as the fourth and second-order singularities in the direct term, which are of comparable magnitude to each other in their respective FSE contributions.

Fourth-order singularity in the direct term. Applying the idea of SS on this term yields the following contribution to the approximate MP2 energy. Note that for brevity, we will drop the second argument, $\mathbf{0}$, for all mentions of S_d as shown in eq (7).

$$E_{\text{MP2}}^{\text{SS}} \leftarrow \frac{1}{N_{\mathbf{k}}} \sum_{\mathbf{q} \in \mathcal{K}_{\mathbf{q}} + \mathbb{L}^*}' \frac{S_d(\mathbf{q}, \mathbf{0}) - A_d(\mathbf{q}, \mathbf{0}; \boldsymbol{\alpha})}{|\mathbf{q}|^4} + \frac{1}{|\Omega^*|} \int_{\mathbb{R}^3} d\mathbf{q} \frac{A_d(\mathbf{q}, \mathbf{0}; \boldsymbol{\alpha})}{|\mathbf{q}|^4}, \quad (13)$$

where the primed sum skips the singular term at $\mathbf{q} = \mathbf{0}$, and $A_d(\mathbf{q}, \mathbf{0}; \boldsymbol{\alpha})$ is the chosen auxiliary function controlled by some parameters $\boldsymbol{\alpha}$ which are fitted to approximate $S_d(\mathbf{q}, \mathbf{0})$ as closely as possible. In particular, the optimal parameters $\boldsymbol{\alpha}_0$ are those that minimize the following loss function:

$$\boldsymbol{\alpha}_0 = \arg \min_{\boldsymbol{\alpha}} \sum_{\mathbf{q} \in \mathcal{K}_{\mathbf{q}} + \mathbb{L}^*, |\mathbf{q}| < q_{\text{cut}}}' \left[\frac{S_d(\mathbf{q}, \mathbf{0}) - A_d(\mathbf{q}, \mathbf{0}; \boldsymbol{\alpha})}{|\mathbf{q}|^4} \right]^2, \quad (14)$$

where q_{cut} is the cutoff radius for the fitting, usually extending multiple Brillouin zones, but including far fewer than the full $N_{\mathbf{k}} N_{\mathbf{G}}$ $\mathbf{q}$ -points in a practical calculation. The choice of loss function is motivated by making the numerical quadrature as small as possible, leading to the smallest error prefactor. Later, after section 3.1, we will assume $A_d(\mathbf{q}, \mathbf{0}; \boldsymbol{\alpha}) = |\mathbf{q}|^4 h_{d,4}(\mathbf{q}; \boldsymbol{\alpha})^2$, where $h_{d,4}(\mathbf{q}; \boldsymbol{\alpha})$ is the decay function of $\mathbf{q}$ used for this term. This allows for the computation of the other corrections to be feasible.

A visualization of the quartic portion of the structure factor $S_d(\mathbf{q}, \mathbf{0})$ and the corresponding fitted auxiliary function $A_d(\mathbf{q}, \mathbf{0}; \boldsymbol{\alpha})$ is shown in Fig.1 for a quasi-1D model system with the same isotropic Gaussian effective potential described in ref [12], except the well depth is raised from -200 to -50 . In this example we use Gaussian decay, $h_{d,4}(\mathbf{q}; \boldsymbol{\alpha}) = \alpha_1 \exp(-\alpha_2 |\mathbf{q}|^2)$.

Second-order singularities in the direct term. These terms are slightly harder to correct, as care needs to be taken to avoid the naive $\mathcal{O}(N_{\mathbf{k}}^3 N_{\mathbf{G}}^2)$ construction of $S_d(\mathbf{q}, \Delta \mathbf{G})$ for all $\mathbf{q}$ and $\Delta \mathbf{G}$ suggested by the last term in eq (7). It is possible to motivate a form for the auxiliary function needed here, as we do later in section 3.1, which suggests $A_d(\mathbf{q}, \Delta \mathbf{G}; \boldsymbol{\alpha}') = |\mathbf{q}|^2 |\mathbf{q} + \Delta \mathbf{G}|^2 h_{d,2}(\mathbf{q}; \boldsymbol{\alpha}') h_{d,2}(\mathbf{q} + \Delta \mathbf{G}; \boldsymbol{\alpha}')$. Here, the

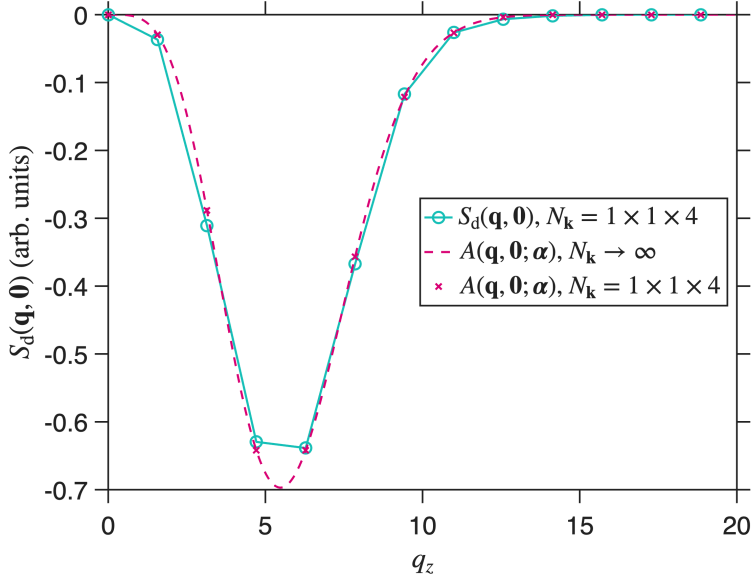


Figure 1: The quartic portion $S_d(\mathbf{q}, \mathbf{0})$ of the MP2 direct term structure factor for the model potential described in the text, plotted along a 1D path in reciprocal space. The fitted quartic auxiliary function $A(\mathbf{q}, \mathbf{0}; \boldsymbol{\alpha})$ is also plotted

decay function $h_{d,2}(\mathbf{q}; \boldsymbol{\alpha}')$ must decay to zero fairly quickly. The function depends on parameters $\boldsymbol{\alpha}'$ which are different from those used for $h_{d,4}(\mathbf{q}; \boldsymbol{\alpha})$.

Taking this idea and applying it along with singularity subtraction yields the following contribution to the MP2 energy

$$E_{\text{MP2}}^{\text{SS}} \leftarrow \frac{1}{N_{\mathbf{k}}} \sum'_{\mathbf{q} \in \mathcal{K}_{\mathbf{q}} + \mathbb{L}^*} \left(\frac{\tilde{S}_{d,2}(\mathbf{q})}{|\mathbf{q}|^2} - \frac{f_{d,2}(\mathbf{q}; \boldsymbol{\alpha}')}{|\mathbf{q}|^2} \right) + \frac{1}{|\Omega^*|} \sum_{\Delta \mathbf{G} \in \mathbb{L}^* \setminus \{\mathbf{0}\}} \int_{\mathbb{R}^3} d\mathbf{q} h_{d,2}(\mathbf{q}; \boldsymbol{\alpha}') h_{d,2}(\mathbf{q} + \Delta \mathbf{G}; \boldsymbol{\alpha}'), \quad (15)$$

where $\tilde{S}_{d,2}(\mathbf{q})$ is the “folded” version of the structure factor $S_d(\mathbf{q}, \Delta \mathbf{G})$ defined as

$$\tilde{S}_{d,2}(\mathbf{q}) := \sum_{\Delta \mathbf{G} \in \mathbb{L}^* \setminus \{\mathbf{0}\}} \frac{S_d(\mathbf{q}, \Delta \mathbf{G})}{|\mathbf{q} + \Delta \mathbf{G}|^2} \quad (16)$$

and $f_{d,2}(\mathbf{q}; \boldsymbol{\alpha}')$ is the corresponding “folded” auxiliary function and defined in terms of $A_d(\mathbf{q}, \Delta \mathbf{G}; \boldsymbol{\alpha}')$ as

$$f_{d,2}(\mathbf{q}; \boldsymbol{\alpha}') := \sum_{\Delta \mathbf{G} \in \mathbb{L}^* \setminus \{\mathbf{0}\}} \frac{A_d(\mathbf{q}, \Delta \mathbf{G}; \boldsymbol{\alpha}')}{|\mathbf{q} + \Delta \mathbf{G}|^2} \quad (17)$$

$$= |\mathbf{q}|^2 h_{d,2}(\mathbf{q}; \boldsymbol{\alpha}') \left[\left(\sum_{\Delta \mathbf{G} \in \mathbb{L}^*} h_{d,2}(\mathbf{q} + \Delta \mathbf{G}; \boldsymbol{\alpha}') \right) - h_{d,2}(\mathbf{q}; \boldsymbol{\alpha}') \right]. \quad (18)$$

In these folded quantities, the sum excludes only the $\Delta \mathbf{G} = \mathbf{0}$ term, which is treated separately by the quartic correction above. Note, firstly, that the singularities at $\mathbf{q} = -\Delta \mathbf{G}$ are removed by the second-order vanishing of $S_d(\mathbf{q}, \Delta \mathbf{G})$ and, secondly, that the quantity in the inner parentheses in eq (18) is periodic with respect to $\mathbf{q}$ in Ω^* . This folding of the structure factor and auxiliary function is an important step that allows efficient calculation of this correction, as it allows us to perform SS and fitting for all $\Delta \mathbf{G}$ terms at once, instead of separately for each $\Delta \mathbf{G}$, a procedure that would have ultimately scaled $\mathcal{O}(N_{\mathbf{k}}^3 N_{\mathbf{G}}^2)$. Furthermore, we do not need to compute $\tilde{S}_{d,2}(\mathbf{q})$ for all possible $\mathbf{q}$, as the very first term in eq (15) is already included in standard MP2. We only need to compute $\tilde{S}_{d,2}(\mathbf{q})$ for as many $\mathbf{q}$ as are needed to obtain a good fit.

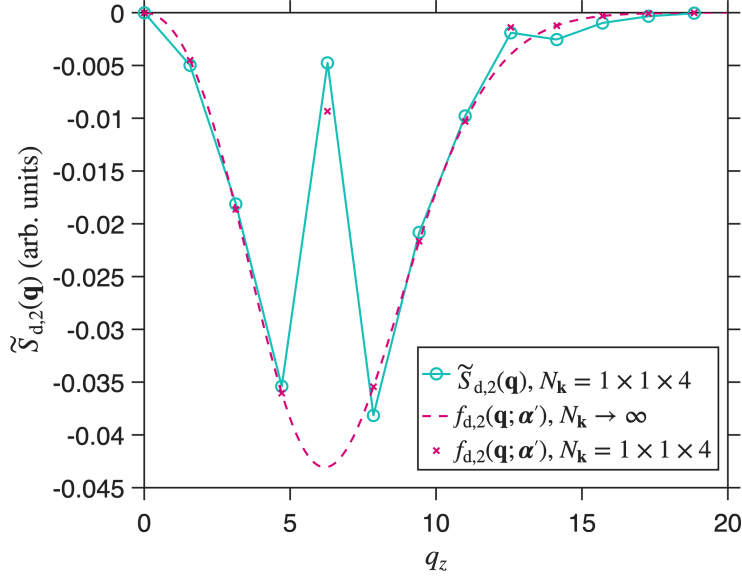


Figure 2: The quadratic portion $\tilde{S}_{d,2}(\mathbf{q})$ of the MP2 direct term structure factor for the model potential described in the text, plotted along a 1D path in reciprocal space. The fitted quadratic auxiliary function $f_{d,2}(\mathbf{q}; \alpha')$ is also plotted, which attempts to capture the behavior of the structure factor including the discontinuities.

Just as in the previous term, the optimal parameters α'_0 are those that minimize the following loss function:

$$\alpha'_0 = \arg \min_{\alpha'} \sum'_{\mathbf{q} \in \mathcal{K}_{\mathbf{q}} + \mathbb{L}^*, |\mathbf{q}| < q_{\text{cut}}} \left[\frac{\tilde{S}_{d,2}(\mathbf{q})}{|\mathbf{q}|^2} - \frac{f_{d,2}(\mathbf{q}; \alpha')}{|\mathbf{q}|^2} \right]^2 \quad (19)$$

where the first term can be computed efficiently for essentially the same reason as for the second term above. The choice of this loss function is a careful tradeoff between minimizing the quadrature term and being computationally feasible (each $\Delta \mathbf{G}$ could be corrected separately, with different parameters, but the scaling of the method would be an unforgiving $\mathcal{O}(N_{\mathbf{G}}^2)$). Finally, an example of the folded structure factor $\tilde{S}_{d,2}(\mathbf{q})$ and the corresponding folded auxiliary function $f_{d,2}(\mathbf{q}; \alpha')$ is shown in fig 2 for the same quasi-1D model system as in the previous section. In this example we also use Gaussian decay, $h_{d,2}(\mathbf{q}; \alpha') = \alpha'_1 \exp(-\alpha'_2 |\mathbf{q}|^2)$.

Exchange Term. Just as in the previous term, it can be motivated (section 3.1) that a good auxiliary function will look like $A_x(\mathbf{q}_1, \mathbf{q}_2; \alpha'') = |\mathbf{q}_1|^2 |\mathbf{q}_2|^2 h_x(\mathbf{q}_1; \alpha'') h_x(\mathbf{q}_2; \alpha'') \sim S_x(\mathbf{q}_1, \mathbf{q}_2)$, where $h_x(\mathbf{q}_1; \alpha'')$ are the now familiar general-form functions fitted with parameters α'' different from those in the previous two subsections.

Using this approximate form for $A_x(\mathbf{q}_1, \mathbf{q}_2; \alpha'')$ and the idea of singularity subtraction, the following final contribution to the MP2 energy can be derived as

$$E_{\text{MP2}}^{\text{SS}} \leftarrow \frac{1}{N_{\mathbf{k}}} \left(\sum'_{\mathbf{q} \in \mathcal{K}_{\mathbf{q}} + \mathbb{L}^*} \frac{\tilde{S}_x(\mathbf{q})}{|\mathbf{q}|^2} - \sum'_{\mathbf{q} \in \mathcal{K}_{\mathbf{q}} + \mathbb{L}^*} \frac{f_x(\mathbf{q}; \alpha'')}{|\mathbf{q}|^2} \right) + \frac{1}{|\Omega^*|^2} \left(\int_{\mathbb{R}^3} d\mathbf{q} h_x(\mathbf{q}; \alpha'') \right)^2. \quad (20)$$

where the first term in the equation is already included in the standard MP2 calculation. Similar to the previous term, the structure factor $\tilde{S}_x(\mathbf{q})$ is the folded version of $S_x(\mathbf{q}_1, \mathbf{q}_2)$ defined as

$$\tilde{S}_x(\mathbf{q}_1) := \frac{1}{N_{\mathbf{k}}} \sum'_{\mathbf{q}_2 \in \mathcal{K}_{\mathbf{q}} + \mathbb{L}^*} \frac{S_x(\mathbf{q}_1, \mathbf{q}_2)}{|\mathbf{q}_2|^2}, \quad (21)$$

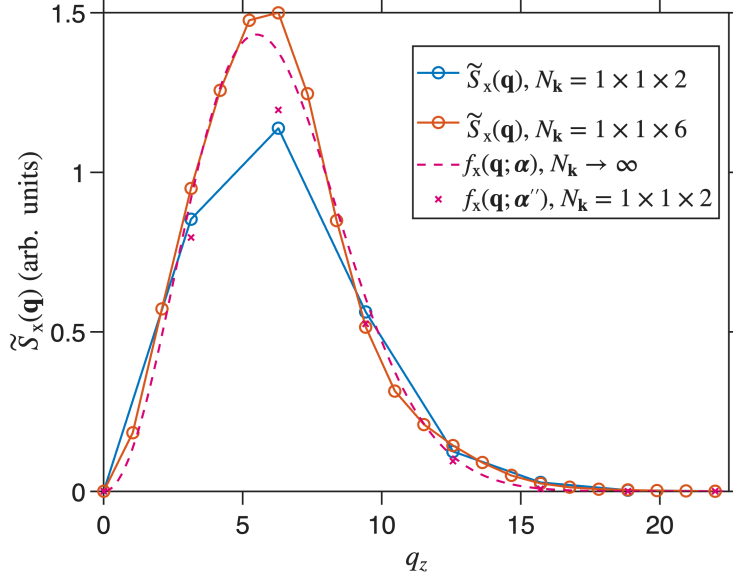


Figure 3: The folded exchange structure factor $\tilde{S}_x(\mathbf{q}_1)$ for the model potential described in the text, plotted along a 1D path in reciprocal space. The fitted folded auxiliary function $f_x(\mathbf{q}_1)$ is also plotted, which attempts to capture the behavior of the structure factor including the discontinuities.

where we have dropped the last two arguments of $S_x(\mathbf{q}_1, \mathbf{q}_2)$, which are implicitly set to $\mathbf{0}$ via eq (10); its corresponding “folded” auxiliary function is defined as

$$f_x(\mathbf{q}_1; \alpha'') := \sum_{\mathbf{q}_2 \in (\mathcal{K}_{\mathbf{q}} + \mathbb{L}^*) \setminus \{\mathbf{0}\}} \frac{A_x(\mathbf{q}_1, \mathbf{q}_2; \alpha'')}{|\mathbf{q}_2|^2} \quad (22)$$

$$= |\mathbf{q}_1|^2 h_x(\mathbf{q}_1; \alpha'') \sum_{\mathbf{q}_2 \in (\mathcal{K}_{\mathbf{q}} + \mathbb{L}^*) \setminus \{\mathbf{0}\}} h_x(\mathbf{q}_2; \alpha''). \quad (23)$$

Once again, this folding allows us to efficiently perform SS for this term, as naively constructing and correcting $S_x(\mathbf{q}_1, \mathbf{q}_2)$ would prohibitively scale $\mathcal{O}(N_{\mathbf{k}}^3 N_{\mathbf{G}}^2)$. The optimal parameters α''_0 are those that minimize the following loss function:

$$\alpha''_0 = \arg \min_{\alpha''} \sum'_{\mathbf{q} \in \mathcal{K}_{\mathbf{q}} + \mathbb{L}^*, |\mathbf{q}| < q_{\text{cut}}} \left[\frac{\tilde{S}_x(\mathbf{q})}{|\mathbf{q}|^2} - \frac{f_x(\mathbf{q}; \alpha'')}{|\mathbf{q}|^2} \right]^2 \quad (24)$$

An example of the folded exchange structure factor and the fitted folded auxiliary function is shown in fig 3 for the same quasi-1D model system as in figs 1 and 2 and with the same Gaussian decay. The $\mathcal{O}(N_{\mathbf{k}}^{-1})$ errors for all $\mathbf{q}$ in $\tilde{S}_x(\mathbf{q})$ are clearly visible when comparing the structure factors at $N_{\mathbf{k}} = 1 \times 1 \times 2$ and $N_{\mathbf{k}} = 1 \times 1 \times 6$. The fitted auxiliary function is able to “reproduce” these discontinuities, which allows for a better approximation to the structure factor at the TDL and thus better FSE corrections.

3.1 Choice of Auxiliary Functions

We have outlined in previous sections how knowledge of the criteria for successful uniform quadrature can be used within a simple correction scheme to produce in theoretically sound and computationally feasible corrections. But these corrections rely on finding auxiliary functions $A_d(\mathbf{q}, \Delta \mathbf{G}; \alpha)$, $A_x(\mathbf{q}_1, \mathbf{q}_2; \alpha)$ that resemble the structure factors $S_d(\mathbf{q}, \Delta \mathbf{G})$, $S_x(\mathbf{q}_1, \mathbf{q}_2)$. In this section we motivate the qualitative shape of the auxiliary functions by considering the symmetries and small $\mathbf{q}$ expansions of the structure factors. The simple form

of the auxiliary functions obtained here also prove crucial in simplifying the expressions and developing a feasible MP2SS scheme (as seen in section 3).

The structure factors, as defined in eqs (6) and (9), have analytic asymptotic expansions. Consider a Taylor expansion of the reciprocal pair densities $\hat{\rho}_{i\mathbf{k}_i,j(\mathbf{k}_i+\mathbf{q})}(\mathbf{G})$ defined in eq (1). When the unfolded wavevector $\mathbf{q} + \mathbf{G}$ approaches $\mathbf{0}$, the zeroth-order term vanishes by orbital orthogonality, so the pair density vanishes linearly in $\mathbf{q} + \mathbf{G}$. Consequently, near the singular limits, $S_d(\mathbf{q}, \mathbf{G}, \mathbf{G}')$ is second order in the small arguments $\mathbf{q} + \mathbf{G}$ and $\mathbf{q} + \mathbf{G}'$, and, similarly, $S_x(\mathbf{q}_1, \mathbf{q}_2, \mathbf{G}_1, \mathbf{G}_2)$ is second order in $\mathbf{q}_1 + \mathbf{G}_1$ and $\mathbf{q}_2 + \mathbf{G}_2$. Taking this small $\mathbf{q}$ behavior and considering the symmetries of the structure factor (section 2.2), it is possible to arrive at multiple expressions that satisfy both symmetry and asymptotics. But perhaps the simplest candidate expressions for the auxiliary functions are

$$A_d(\mathbf{q}, \Delta\mathbf{G}) = |\mathbf{q}|^2 |\mathbf{q} + \Delta\mathbf{G}|^2 h_d(\mathbf{q}) h_d(\mathbf{q} + \Delta\mathbf{G}) \quad (25)$$

$$A_x(\mathbf{q}_1, \mathbf{q}_2) = |\mathbf{q}_1|^2 |\mathbf{q}_2|^2 h_x(\mathbf{q}_1) h_x(\mathbf{q}_2) \quad (26)$$

where decay functions h_d, h_x have been introduced. The prefactors to these decay functions capture the correct asymptotic behavior, while the decay functions ensure a smooth decay of the auxiliary functions at large $\mathbf{q}$.

The theoretical guarantees of SS allow for imperfect auxiliary functions to still improve the rate of error convergence. But a good auxiliary function that accurately captures the structure factors will lead to an additional advantage: a small error prefactor on top of the improved convergence rate. For this reason, we give more flexibility to our auxiliary functions by furnishing the decay functions with optimizable parameters α , represented explicitly as $h_d(\mathbf{q}; \alpha)$, $h_x(\mathbf{q}; \alpha)$, as done previously [18].

Due to the complexity of real systems, which prevents acquiring concise closed-form expressions for the structure factors, the choice of decay functions must be done mostly empirically. Here, we present three possible configurations of MP2SS, each with their specific choice of auxiliary function decay, as outlined in table 1. All auxiliary functions across all configurations are fitted with the appropriate Coulomb-weighting for each term, as seen in eqs (14), (19) and (24).

The first two configurations, MP2SS-Gaussian and MP2SS-Exponential, are the simplest, and use the same decay function for all terms. Gaussian decay reflects both the property that the direct and exchange structure factors at the thermodynamic limit are smooth as well as the assumption that, because we are using Gaussian basis sets, should decay like Gaussians at large $\mathbf{q}$. Exponential decay, on the other hand, allows for more better agreement in the behavior of the auxiliary function near the origin at the cost of losing smoothness. Previous work [18] has also shown that exponential decay may be appropriate for larger Gaussian basis sets, which can better capture non-smooth features like cusps and therefore result in structure factors which decay slower than a Gaussian.

We also present a third configuration, MP2SS-Tuned, which uses different decay functions for each term to optimize the FSE curves over our test set. It is possible to justify the choices as follows. As observed in other work in FSE corrections for post-SCF methods [16], very fine $\mathbf{k}$ -meshes are needed to resolve the smooth part of the structure factor at small $\mathbf{q}$ in real materials. In our work, we find the fourth-order direct term specially suffers from this. Since in our usual computational $\mathbf{k}$ -meshes the smoothness of this term is not resolved, and it appears to have a cusp, we choose the exponential decay function to fit it. The second-order direct term is better resolved, so its smooth part is visible and a smooth auxiliary function is best used. Since the flexibility of a gaussian seems enough to capture the shape of this term, we decide to not introduce a more complex auxiliary function. Finally, it seems the exchange term benefited from a slightly more flexible auxiliary function, so the quartic exponential function was introduced, mirroring our previous work [18]. This MP2SS-Tuned configuration in fact exhibits the best performance across the board for our test set.

The 1-particle basis also has a significant impact on the structure factor. For example, if a basis set is unable to capture the correct physical phenomena in the system (as is the case for systems with dispersive interactions in basis sets without polarisation), the structure factor will not be converged, and it will look qualitatively differently than at the complete basis set limit. In general, we observe that there is a qualitative change of shape in the structure factor as the basis set is increased. This means that a particular auxiliary function will perform differently in different basis sets. In this paper, our objective is to be working as close as to the MP2 basis-set limit as is computationally feasible. For a further discussion on the effect of the basis set on the structure factor and the performance of MP2SS, see Section S1 in the SI.

Table 1: Summary of the decay functions used for the auxiliary functions for each MP2 term. The functions $h_{d,4}(\mathbf{q}; \alpha)$, $h_{d,2}(\mathbf{q}; \alpha')$, and $h_x(\mathbf{q}; \alpha'')$ correspond to the direct term 4th order singularity, direct term 2nd order singularity, and exchange term respectively, and they are all functions of $\mathbf{q}$ only through its magnitude $q = |\mathbf{q}|$. The total number of optimizable parameters are 2 or 4 depending on the function, and they are labeled α_i . Each column represents a different collection of function choices making up one of the three tested configurations: MP2SS-Gaussian, MP2SS-Exponential or MP2SS-Tuned.

Decay Function	Gaussian Config.	Exponential Config.	Tuned Config.
$h_{d,4}(\mathbf{q}; \alpha)$	$\alpha_1 \exp\left(-\frac{q^2}{2\alpha_2^2}\right)$	$\alpha_1 \exp(-\alpha_2 q)$	$\alpha_1 \exp(-\alpha_2 q)$
$h_{d,2}(\mathbf{q}; \alpha')$	$\alpha_1 \exp\left(-\frac{q^2}{2\alpha_2^2}\right)$	$\alpha_1 \exp(-\alpha_2 q)$	$\alpha_1 \exp\left(-\frac{q^2}{2\alpha_2^2}\right)$
$h_x(\mathbf{q}; \alpha'')$	$\alpha_1 \exp\left(-\frac{q^2}{2\alpha_2^2}\right)$	$\alpha_1 \exp(-\alpha_2 q)$	$\alpha_1 \exp(-\alpha_2(\sqrt{1 + \alpha_3 q^2 + \alpha_4 q^4} - 1))$

3.2 Computational Procedure

After the choice of MP2SS configuration is made, the procedure to obtain the MP2SS correction is as follows: (1) perform a periodic HF calculation to obtain the orbitals and orbital energies; (2) compute the exact exchange structure factors for each term and fit the ExxSS auxiliary function to obtain the ExxSS correction to the occupied orbital energies (obtained by dividing the ExxSS correction by N_{occ}); (3) perform a periodic MP2 calculation with the ExxSS-corrected orbital energies to obtain the MP2 structure factor; (4) fit the MP2SS auxiliary function to the MP2 structure factors via eqs (14), (19) and (24) and evaluate the integral of the auxiliary function to obtain the MP2SS correction. A flowchart of this procedure is shown in Figure 4. For ExxSS, we follow the recommendations outlined in our previous work [18] for the choice of auxiliary function (quartic-exponential) and fitting procedure. Numerical experiments on insulators (see figures S4 and S5 in the SI) show that the corrections provided by ExxSS to the MP2 correlation energy are small compared to MP2SS, indicating that for such systems, ExxSS may not be necessary to achieve good convergence toward the TDL. ExxSS, however, is not the bottleneck in MP2SS.

The walltime bottleneck of the MP2SS procedure is the building of the MP2 structure factors (for the purpose of fitting the auxiliary functions), scaling up to $\mathcal{O}(N_{\mathbf{k}}^3 N_{\mathbf{G}} N_{\text{occ}}^2 N_{\text{virt}}^2)$; specifically, computing the structure factor at each $\mathbf{q} + \mathbf{G}$ point sampled scales $\mathcal{O}(N_{\mathbf{k}}^2 N_{\text{occ}}^2 N_{\text{virt}}^2)$, and the number of $\mathbf{q} + \mathbf{G}$ points scales at worst $\mathcal{O}(N_{\mathbf{k}} N_{\mathbf{G}})$, although a cutoff may be used to reduce the number of points sampled as discussed in the previous section. To accelerate this portion, we only sample the points within a maximum q radius, which is set to be twice that of the ExxSS structure factor fit radius, again following the recommendations from our previous work [18]. Furthermore, we construct the pair densities in eq 1 via a coarse real-space quadrature corresponding to a kinetic energy cutoff of 30 hartree. Results are relatively insensitive to the choice of q radius cutoff, and readers are directed to figure S11 in the SI for convergence tests.

We can also further reduce the runtime by sampling the structure factor only along the reciprocal lattice vectors instead of within a sphere of a certain cutoff radius. We denote all MP2SS configurations that use this sampling method with the “-LS” (line-sampling) ending, and the reduced sampling allows us to set the maximum q for which we sample to be three times that of the ExxSS procedure. Line-sampling, while not providing as good a fit to the structure factor in principle, still allows us to fit to the diagonal terms of the Taylor expansion of the structure factors at $\mathbf{q} + \mathbf{G} = \mathbf{0}$. As shown in the results for diamond at the triple zeta level in figure 9, the FSE correction is not significantly affected compared to when sampling all $\mathbf{q} + \mathbf{G}$ within a ball. While the a rigorous optimization of the runtime is not the primary focus of this work, readers are directed to section S4 in the SI for preliminary benchmarks of SS versus the staggered mesh and uncorrected MP2 routines that achieve similar levels of accuracy toward the estimated TDL.

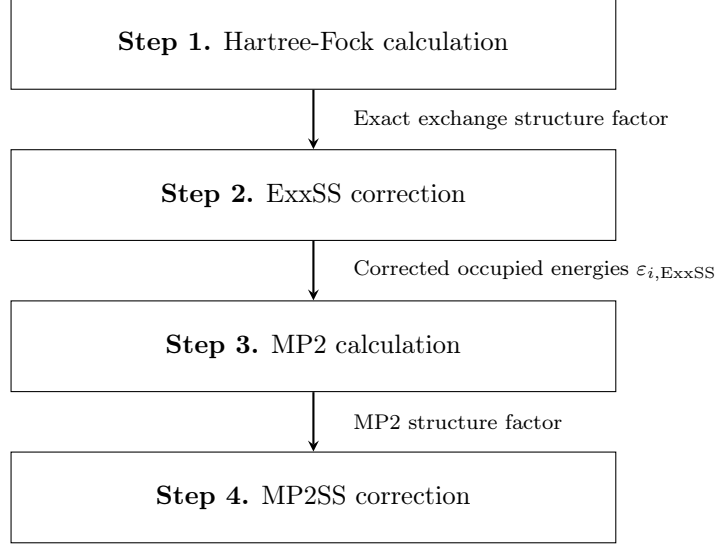


Figure 4: High-level flowchart to obtain the singularity subtracted MP2 correlation energy.

4 Results

We now show the effectiveness of MP2SS for reducing the FSE in the canonical MP2 correlation energy for two solids and two molecular crystals: diamond, silicon, ammonia, and urea. The unit cells for diamond and silicon are obtained from the Materials Project, which uses the PBE functional [25] for geometry optimization, and the unit cells for ammonia and urea are obtained from the X23 dataset [26, 27], which also uses the PBE functional with the TS dispersion correction [28]. All calculations were performed with the PySCF package [29, 30], using a kinetic energy cutoff of 56 hartree and the GTH-PBE pseudopotential and basis set [31]. Unless otherwise stated, all calculations were performed with PySCF default Gaussian Density Fitting (GDF) approach [29, 32]. We use Monkhorst-Pack (MP) $\mathbf{k}$ -point grids that are Γ -centered for odd $\mathbf{k}$ -meshes and shifted by half a grid-space in each direction to sidestep the Γ -point for even $\mathbf{k}$ -meshes. As already mentioned, our test materials were limited to insulators and semiconductors, as MP2 diverges for metals. For each FSE curve, we define a reference TDL by fitting the uncorrected data to the leading-order form $E(N_{\mathbf{k}}) = aN_{\mathbf{k}}^{-1} + b$ and taking b as the extrapolated intercept. We indicate when the data is within millihartree accuracy per atom/monomer of this reference with a gray stripe. Because these sparse $\mathbf{k}$ -meshes might not yet be fully in the asymptotic regime, this extrapolated value should be interpreted as a practical benchmark, and log-log error plots are not presented. Regular FSE curves already highlight which methods perform well. Nonetheless, readers can consult figure S12 in the SI for an example log-log error plot.

We observed in our numerical experiments that the staggered mesh method appeared to approach a different limiting value than the other methods, especially for molecular crystals. This is most likely because the staggered mesh method implementation in PySCF used here computes the density matrix and orbital energies at both the shifted and unshifted $\mathbf{k}$ -points using a non-SCF band structure calculation that involves the construction of the Hartree operator J and the exact-exchange operator K via a truncated Coulomb potential. This truncated Coulomb kernel is only available with Fast Fourier Transform density fitting (FFTDF) in PySCF at the time of writing. Therefore, staggered mesh uses GDF for the two-electron integrals calculation and FFTDF for the density matrix, as opposed to all other results that use *only* GDF. This causes staggered mesh to exhibit a slightly different apparent limit if the auxiliary basis is not large enough. Thus, for all results shown below (except for diamond with the GTH-SZV basis), the staggered mesh curve is shifted by a constant offset so that its densest available $\mathbf{k}$ -mesh value coincides with the extrapolated all-GDF reference. This alignment is used only to compare the decay of the FSE; it should not be interpreted as showing agreement in the absolute energies or in the true limiting value. For the unshifted curves, readers are directed to section S2 in the SI. Based on those unshifted curves, the difference between the apparent limiting values of the staggered-mesh and all-GDF data appears to be at most a few

millihartree per atom/monomer.

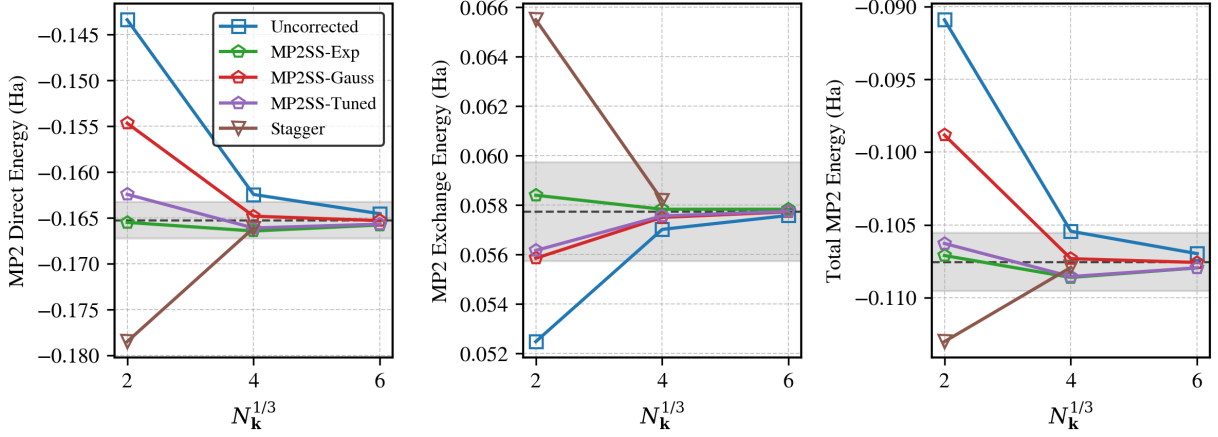


Figure 5: Finite-size effect of the direct, exchange, and total MP2 correlation energy per unit cell with the GTH-SZV basis set for diamond with different MP2SS configurations. The results for the (shifted) staggered mesh method are also included for comparison. The gray horizontal stripe represents millihartree accuracy per atom with respect to the fitted TDL reference obtained from an $N_{\mathbf{k}}^{-1}$ fit to the uncorrected curve. The $N_{\mathbf{k}} = 6 \times 6 \times 6$ point for staggered mesh is missing due to disk space constraints from building the GDF Cholesky vectors.

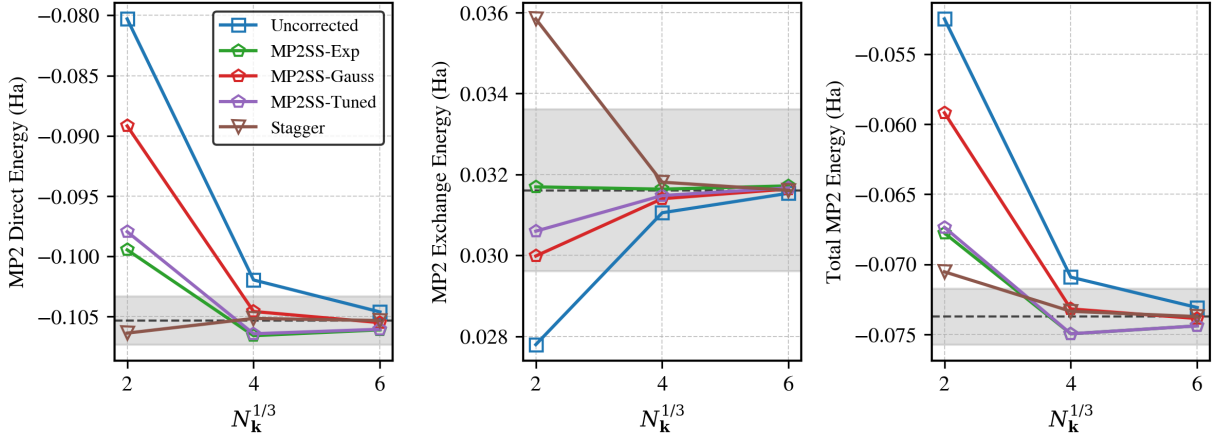


Figure 6: Finite-size effect of the direct, exchange, and total MP2 correlation energy per unit cell with the GTH-SZV basis set for silicon with different MP2SS configurations. The gray horizontal stripe represents millihartree accuracy per atom with respect to the fitted TDL reference obtained from an $N_{\mathbf{k}}^{-1}$ fit to the uncorrected curve.

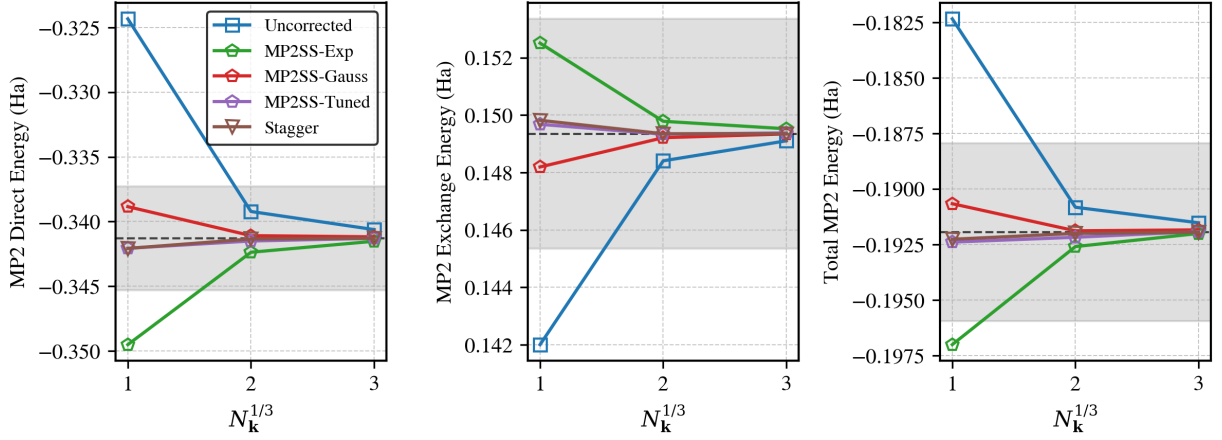


Figure 7: Finite-size effect of the direct, exchange, and total MP2 correlation energy per unit cell with the GTH-SZV basis set for ammonia with different MP2SS configurations. The gray horizontal stripe represents millihartree accuracy per monomer with respect to the fitted TDL reference obtained from an N_k^{-1} fit to the uncorrected curve.

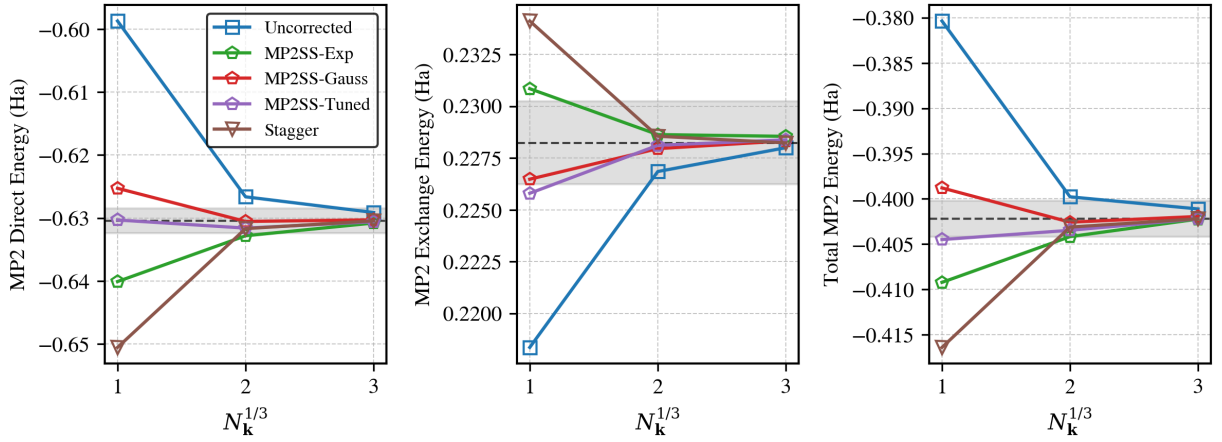


Figure 8: Finite-size effect of the direct, exchange, and total MP2 correlation energy per unit cell with the GTH-SZV basis set for urea with different MP2SS configurations. The gray horizontal stripe represents millihartree accuracy per monomer with respect to the fitted TDL reference obtained from an N_k^{-1} fit to the uncorrected curve.

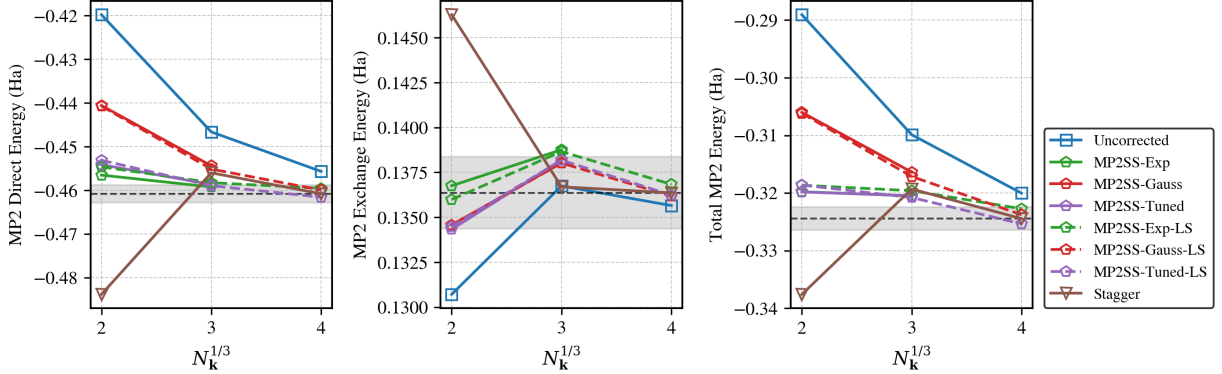


Figure 9: Finite-size effect of the direct, exchange, and total MP2 correlation energy per unit cell with the GTH-cc-pVTZ basis set for diamond with different MP2SS configurations. The gray horizontal stripe represents millihartree accuracy per atom with respect to the fitted TDL reference obtained from an $N_{\mathbf{k}}^{-1}$ fit to the uncorrected curve. The data point at $N_{\mathbf{k}} = 3^3$ is not included when constructing this reference. The dashed curves represent MP2SS corrections when the auxiliary function is fit to line-sampled (LS) structure factor data. See text for details.

We first discuss the performance of the three MP2SS configurations for the GTH-SZV basis set. Within this test set, the MP2SS-Tuned configuration performs the best out of all configurations, achieving millihartree accuracy per atom or monomer with respect to the total correlation energy at the fitted TDL reference at the coarsest $\mathbf{k}$ -meshes before any other configuration. MP2SS-Gauss performed the worst with the smaller unit cells (diamond and silicon), but performed better on the molecular crystals. The results were the opposite for MP2SS-Exponential, which performed better with diamond and silicon but often provided an overcorrection for the molecular crystals. The respective performances of MP2SS-Gaussian and MP2SS-Exponential across this test set may be rationalized by the role that the decay function plays in the small- q behavior of the Coulomb-weighted quantities entering the loss functions. A Gaussian decay is an even, smooth function of q , whereas $\exp(-\alpha q)$ has a cusp at $q = 0$ and contains both even and odd powers in its one-sided expansion. The exponential form therefore offers additional flexibility near the origin, but on systems whose weighted structure factors are effectively smooth and even at the resolution of the sampled meshes it can also be more prone to overcorrection. In contrast to MP2SS-Gaussian and MP2SS-Exponential, the MP2SS-Tuned configuration uses decay functions and fitting procedures that are chosen for each MP2 term based on how well they correct that specific term rather than *a priori* estimations of how the auxiliary function should behave near the origin. Nonetheless, it outperforms the other two configurations, which suggests that granularity in how each singularity is treated is important for the performance of singularity subtraction in general.

When we move to the larger GTH-cc-pVTZ basis set [33] for diamond, we also include results from the line-sampling of the structure factor to allow us to compute the correction at $N_{\mathbf{k}} = 4 \times 4 \times 4$. We first observe that results do not change significantly when sampling the structure factor along the reciprocal lattice vectors instead of all $\mathbf{q} + \mathbf{G}$ within a certain radius, indicating that not all $\mathbf{q} + \mathbf{G}$ points are required to be sampled in order to obtain a good auxiliary function fit. We also observe that MP2SS-Tuned and MP2SS-Exponential perform better than MP2SS-Gaussian, which suggests that for larger basis sets and smaller unit cells, the smoothness of the structure factor near the origin is not fully resolved. This general motif of not being able to resolve the structure factor behavior near the origin has also been previously observed in other periodic post-SCF calculations [16]. Conversely, for larger unit cells and smaller basis sets, the smoothness of the structure factor near the origin is better sampled, and the selection of the MP2SS-Gaussian configuration becomes more appropriate.

Finally, we note that in most cases, MP2SS (especially MP2SS-Tuned) is closer to the TDL than staggered mesh at the coarser $\mathbf{k}$ -meshes, and has similar performance at denser $\mathbf{k}$ -meshes. The main exception is silicon, which has the smallest bandgap in our set and has a structure factor whose behavior around $\mathbf{q} = \mathbf{0}$ is not

resolved at sparser $\mathbf{k}$ -meshes.

5 Discussion

This work demonstrates that the singularity subtraction approach—previously shown to be effective for exact exchange in periodic HF and hybrid DFT calculations—can be extended to MP2 calculations on real systems. By explicitly accounting for the singularities and discontinuities in the correlation energy structure factor through carefully chosen auxiliary functions, MP2SS achieves significantly faster convergence to the thermodynamic limit compared to uncorrected calculations and competes with the staggered mesh method without the need for additional density and orbital energy calculations at shifted $\mathbf{k}$ -meshes. The folded auxiliary function approach introduced in this work allows SS to directly address the discontinuities in the MP2 structure factor and obtain better fit parameters and provide accurate corrections even when individual structure factor components are not well-resolved at finite $N_{\mathbf{k}}$.

Among the three MP2SS configurations tested, MP2SS-Tuned consistently delivers the best performance across our test set of solids and molecular crystals, often achieving millihartree accuracy at coarse $\mathbf{k}$ -point meshes. The superior performance of MP2SS-Tuned suggests that tailoring a strategy to each specific singularity is more effective than applying a uniform approach across all terms. The MP2SS-Gaussian and MP2SS-Exponential configurations, while simpler and more consistent in their choice of decay functions, exhibit performance that depends on the system and basis set. MP2SS-Gaussian performs better for larger unit cells and smaller basis sets, while MP2SS-Exponential is better for smaller unit cells and larger basis sets.

Further improvements can still be made to the MP2SS methodology. First, the ability to directly compute the first few derivatives of the structure factor at the origin could enhance the accuracy of the auxiliary-function fit, particularly for systems where this small- $\mathbf{q}$ expansion is not resolved by the accessible $\mathbf{k}$ -meshes, as expected for low band-gap materials such as silicon. Initial numerical tests suggest that the performance of MP2SS improves if small- $\mathbf{q}$ values of the structure factor could be reliably sampled. Second, as briefly mentioned in the results and computational procedure sections, we could also improve how the structure factor is built and sampled. The quadrature corrections and auxiliary-function integrals in eq (12) are negligible in all of the above calculations; the dominant additional cost comes instead from constructing the MP2 structure-factor data used in the fits.

The singularity subtraction framework developed here is naturally extensible to other post-SCF theories such as MP3 and coupled cluster, which exhibit similar singularities in their correlation energy expressions [13, 15]. Local and regularized MP2 methods, such as size-consistent Brillouin-Wigner perturbation theory (BW-s2)[34], also stand to benefit from this approach. Because MP2 diverges for metals and performs poorly for low band-gap materials, BW-s2 with the SS scheme in particular presents a promising way to accurately calculate the correlation energy for these challenging systems.

Extensive and systematic benchmarking of more materials at larger basis sets would further highlight MP2SS’s performance. In particular, these promising results suggest MP2SS could perform well in obtaining MP2 cohesive energies at the TDL for molecular crystals using only Γ -point calculations [6]. As aforementioned, we also aim to extend this work to metals and more semiconductors with regularized MP theories.

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Supporting information

Discussion on the basis set dependence of the MP2 structure factor. FSE curves without the shift for the staggered-mesh correction. Second and fourth order contributions to the direct term correction. Preliminary walltime benchmarks. $\mathbf{q}$ -cut convergence plots and log-log error plot for diamond. FSE curves for diamond and silicon which include the effect of only using ExxSS to correct the orbital energies. Raw data used to generate the FSE curves. This information is available free of charge via the Internet at <http://pubs.acs.org>.

Data and Software Availability

The data underlying this study are available in the published article and in the Supporting Information, which includes the data used to plot the finite-size error curves in CSV format. The MP2SS code is also available as an installable Python package at <https://github.com/stephen-quinton/fsec>, along with example calculations.

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MP2 k-space Singularities

$$\frac{1}{|\Omega^*|} \int_{\mathbb{R}^3} d\mathbf{q} \frac{S_d(\mathbf{q}, \mathbf{0}, \mathbf{0})}{|\mathbf{q}|^4}$$

$$\frac{1}{|\Omega^*|} \int_{\mathbb{R}^3} d\mathbf{q} \sum_{\Delta \mathbf{G} \neq \mathbf{0}} \frac{S_d(\mathbf{q}, \mathbf{0}, \Delta \mathbf{G})}{|\mathbf{q}|^2 |\mathbf{q} + \Delta \mathbf{G}|^2}$$

$$\frac{1}{|\Omega^*|} \int_{\mathbb{R}^3} d\mathbf{q}_1 \int_{\mathbb{R}^3} d\mathbf{q}_2 \frac{S_s(\mathbf{q}_1, \mathbf{q}_2, \mathbf{0}, \mathbf{0})}{|\mathbf{q}_1|^2 |\mathbf{q}_2|^2}$$

Singularity Subtraction

